Haoran (Howard) Ding

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EDUCATION

University of California, Los Angeles **Los Angeles, CA**
B.S. Electrical Engineering, Samueli School of Engineering | GPA: 3.74/4.00 *Expected June 2029*
Relevant Coursework: CS 31 & CS 32 (Programming in C++ I & II), Math 33A (Linear Algebra), Math 33B (Differential Equations), CS M51A (Logic Design of Digital Systems), ECE 102 (Systems and Signals)

RESEARCH EXPERIENCE

UCLA Summer Undergraduate Research Program (SURP) **Los Angeles, CA**
Research Fellow — Terahertz and Infrared Photonics Group *June 2026 – Present*
- Replaced the lab's manual LabVIEW test configuration with a MATLAB automation suite that controls and sequences a multi-instrument setup (step-scan FTIR spectrometer, function generator, lock-in amplifier) over GPIB.
- Wrote scripts to automate instrument communication and full device characterization sequences — biasing the device under test and capturing output across pulsed and continuous-wave operating conditions — eliminating manual reconfiguration and improving test repeatability.
- Wrote data analysis scripts to process acquired test data, generate performance plots, and extract key device parameters (threshold current density, slope efficiency, characteristic temperature) against expected operating behavior.
- Built a MATLAB GUI letting lab members configure test parameters and run standardized validation sequences without editing code; deployed to the shared lab PC via a self-updating Git distribution.
- Architected modular instrument drivers (SCPI/serial/TCP transports) for the Keithley 2420 source meter, SR830 lock-in amplifier, Nicolet 8700 FTIR spectrometer, and 8114A pulse generator, with mock instruments enabling full GUI validation without hardware attached.
- Independently brought up and characterized a previously untested hardware device — a metal-metal waveguide terahertz quantum cascade laser — establishing its full operating envelope: 2.45–2.61 THz bias tuning range and 133 K maximum operating temperature.

EXPERIENCE

Handshake AI **Remote**
AI Training Fellow *January 2026 – Present*
- Evaluate and refine frontier Large Language Models via structured Reinforcement Learning from Human Feedback (RLHF) pipelines, improving model reasoning quality and output accuracy.
- Audit complex AI-generated technical solutions for logical consistency across engineering and CS domains, providing calibrated feedback that informs model fine-tuning.

Bruin Underwater Robotics (ASME) at UCLA **Los Angeles, CA**
Robotics Engineer — Electrical Subteam *September 2025 – Present*
- Responsible for the AUV battery management system (BMS) board's step-down subsystem, converting the 14.4V battery rail to a regulated 9V rail; the current revision uses an LDO, and I am developing a buck converter replacement to improve conversion and thermal efficiency.
- Contributed to AUV chassis frame architecture and thruster configuration as part of the mechanical design team, evaluating trade-offs in rigidity, buoyancy distribution, and hydrodynamic stability while collaborating with the electrical subsystem team on component integration.

MATE ROV Underwater Robotics **San Jose, CA**
Team CEO *September 2022 – May 2025*
- Led a multidisciplinary team through the full design-build-pilot lifecycle of a Remotely Operated Vehicle, earning 1st Place Overall and 1st Place in Engineering & Communication at the 2023 MATE ROV Monterey Regional.
- Spearheaded CAD development of critical ROV components in Onshape, applying hydrodynamic design principles and iterative 3D-printing prototyping to minimize structural weight and improve efficiency.

PROJECTS

AI-Powered Market Analyst Agent **2026**
Python | Gemini API | Financial Modeling Prep API
- Built a Gemini-powered agent that researches a public company (fundamentals, DCF, 10-K risk factors, earnings calls, peer comps) and writes a schema-validated Buy/Hold/Sell memo, with guardrails that reroute banks and insurers from DCF to P/B + ROE valuation.
- Shipped it as a FastAPI + React/TypeScript app with live SSE streaming of the agent's tool calls and a 149-test suite (120 Python, 29 frontend); each analysis completes in 1–3 minutes versus hours of manual research.

Quantum RAG Assistant **2026**
Python | ChromaDB | LLM Integration | React/TypeScript | LangGraph
- Built an agentic RAG system over THz photonics papers to accelerate literature review for my lab research: a LangGraph ReAct agent whose grader node re-retrieves with reformulated queries when answers aren't grounded, returning answers cited to source PDF page numbers.
- Productionized it as a FastAPI + React/Vite app with API-key auth, per-IP rate limiting, CORS, and per-thread SQLite checkpointer memory for multi-turn conversation; CI runs backend/frontend tests and deploys the frontend.

Autonomous Line-Following Robot **2026**
C++ | Arduino | PID Control
- Built a line-following robot on Arduino (C++) using an 8-sensor IR array with per-sensor calibration and weighted-error fusion driving dual motor drivers; tuned the PID loop to minimize tracking error and oscillation across variable-contrast surfaces, adding track-loss recovery and finish-line detection.

TECHNICAL SKILLS

Languages: MATLAB, Python, C++, C, Java, git, TypeScript
Instrumentation: Instrument Control & Automation, FTIR Spectroscopy, Lock-In Amplifiers, Function & Pulse Generators, Cryostats, LabVIEW, MATLAB GUI Development, GPIB communication, NI-VISA, oscilloscopes
AI & Software: Gemini API, RAG, ChromaDB, RLHF, React, FastAPI, Streamlit, REST API Design, CI/CD, LangGraph, Vitest, React Testing Library
Hardware & Design: Arduino, STM32, Circuit Analysis, Soldering, CAD, 3D Printing